



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Understanding cybernetics and the systems theory



Learning Outcome:

Understands systems theories, business processes and basic concepts on IT.

Create a map of processes that allows identifying areas in an information system, using IT tools to generate reports based on the company's needs.

Ask your partner:

- How do systems coordinate actions to get the desired results?
- Do you think cybernetic theory is an advance for society? Why, or why not?

Can Do Statements:

- I can give information about systems theory definition.
- I can list examples of systems theory applications.
- I can talk about the relationship between cybernetics and systems theory.



Let's Explore!

Systems and Theories!



Instructions: Read the text below.

Is it important for software engineers to know how systems work and interact together? Absolutely, they need to know systems theory because it is the analysis of complex systems' interactions in nature, society, and science. It provides information about systems' structures and the description of particular groups, their work, and results. And those groups can be electro-mechanical or informational artifacts. They can also be used to identify two types of systems: closed systems, for groups that have interactions solely with members of a group; and open systems, for groups that have external relations and constant environmental changes. The main functions of an information system are: input methods, for collecting group information and data processing, and output methods, which are a way of transferring information.

Is systems theory practical in its application? It's a question worth considering. There are different examples about the ways you can apply systems theory. In informational systems, for example, technological devices allow you to record, store, and write different codes. There are also examples in the application of systems theory in companies, which use it to make decisions, infer problems, control operations, and produce inventions. That's why systems theory models are an integral part of most business organizations.

Now, are computer science and business important to informational systems? Yes, they are. Systems theory is, in fact, an interdisciplinary model. So, it integrates knowledge from computer science and business organizations to demonstrate group dynamics. Many authors use cybernetics and systems theory as the same concept. However, cybernetics is a way to provide feedback and control to artificial systems in technology, and systems theory uses this to understand systems' interaction. As a result, is it possible to consider cybernetics as a term that complements systems theory? Yes, cybernetics is indeed a complement to extensive systems theory.

**Let's Practice!**

Learning what system theory is about.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

Are / Is / Is

1. _____ systems theory the analysis of complex systems interaction?
2. _____ systems theory the description of a particular group work and results?
3. _____ electro-mechanical and informational artifacts part of the systems theory group?

**Let's Practice!**

Systems theory can be very useful for everyone.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: can be, can represent, can record.

1. Systems theory _____ very practical in application
2. With systems theory you _____, store and spread different codes.
3. Systems theory _____ business organizations philosophy.





Extension Activity

A list about software for companies

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about software for companies.
- Describe at least five of those software.
- Write how that software help the companies taking into account system theory.
- Share your list with the class and discuss about more software and their uses.

Self-Assessment

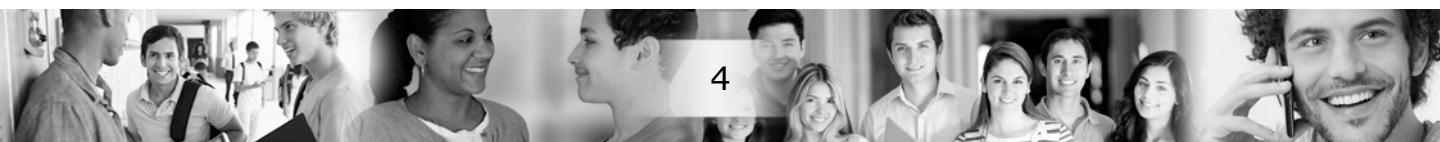
Yes **No**

- | | | |
|---|-----------------------|-----------------------|
| 1. I find information about software for companies. | ☐ | ☐ |
| 2. I describe software functions. | ☐ | ☐ |
| 3. I understand the importance of software for companies. | ☐ | ☐ |
| 4. I connect software with system theory. | ☐ | ☐ |



Target Words

- **in fact** (adverb) it is used to talk about something that is really true
e.g. System theory is in fact, an interdisciplinary model.
- **cybernetics** (noun) cybernetics is a theory for feedback and control of mechanic and artificial systems in technology
e.g. Many authors use cybernetics and systems theory as the same concept.
- **output** (noun) an intentional response or result produced by a system
e.g. The main functions are: inputs to collect group information, data processing and outputs.
- **electro-mechanical** (adjective) it refers to mechanical devices that work thanks to electric systems
e.g. Those groups can be electro-mechanical or informational artifacts.
- **input** (noun) a stimulus, information to be entered into an entity for processing
e.g. The main functions of an information system are: inputs to collect group information.
- **worth** (adjective) it refers to something of a great value
e.g. It's a question worth considering.





Learning Points

Simple Present Tense - Verb To Be Interrogative Form

You use **simple present** tense to talk about general actions, habits and facts about any subject.

You can use the following structure to ask questions in Simple Present tense with the **verb to be (am, is, are)**:

To be (am, is or are) + subject + verb (base form) + complement+?

Example:

Is the systems theory a support for human sciences?

Am I right when I say that cybernetics is an advance?

Remember that in English, questions do not have a question mark at the beginning. Check the corresponding conjugation:

- Am I right?
- Is he here?
- Is she an engineer?
- Is it an artificial system?
- Are we wrong?
- Are you software students?
- Are they scientist or engineers?

Examples

- *Is systems theory practical in application?*
- *Are computer science and business important to information systems?*
- *Is it possible to consider cybernetics as a term that complements systems theory?*
- *Is it important for software engineers to know how systems work and interact together?*





Learning Points

Cybernetics and Systems Theory

Cybernetics and Systems Theory are two controversial topics in information systems and computer science.

Cybernetics	Systems Theory
It is an approach mostly used in science communication to understand the control of machines. Cybernetics permits communication channels to optimize the diffusion of information. The cybernetic approach considers an engineering control concept: "feedback loop" which is to send back the information from the output to the input of the system and, according to the system reaction, it is a negative or an affirmative feedback. When it is a negative feedback, the system is stable. On the contrary, when it is positive, there is an exponential growth or decline.	It is an interdisciplinary approach because it involves mathematics, technology, biology and some other social sciences. Systems theory states how it is possible to find different types of organizations thanks to the variety in systems. It focuses on different connections and interactions of system components. In theory it emphasizes all types of systems; however, in practice it contemplates complex, flexible, and autonomous systems such as cybernetic.
DIFFERENCE	
Cybernetics ↓ How systems <u>function</u> ↓ How they control their actions ↓ How they communicate with other systems and their other components	Systems Theory ↓ Studies: ↓ The <u>structure</u> of systems ↓ Systems types and models
Cybernetics and Systems Theory are two aspects of the same approach because the structure and function of a system cannot be divided.	